

Comprehensive Engineering Analysis and Implementation of a Dual-Axis Automated Paper Fabrication and Shearing System Using STM32 Microcontroller Architecture

A Project Report

Prepared by

Priyanku Buragoin

3rd Year Student

Department of Electronics and Communication Engineering (ECE)

Abstract

The integration of embedded systems with precision mechanical actuators has redefined the capabilities of small-scale industrial automation. This project presents the design and implementation of an automated paper-cutting machine that combines mechanical design, power electronics, and embedded software using the STM32F103 microcontroller — a 32-bit ARM Cortex-M3 based platform capable of high-speed processing and precise peripheral control. The system employs a vertical lead-screw clamping mechanism and a horizontal rack-and-pinion shearing assembly, coordinated through sensorless current-based feedback and dedicated motor-driving circuitry, achieving a level of repeatability and precision difficult to attain through manual operation.

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The integration of embedded systems with precision mechanical actuators has redefined the capabilities of small-scale industrial automation. The development of an automated paper-cutting machine requires a sophisticated synthesis of mechanical design, power electronics, and high-level software logic. By utilizing the STM32F103 microcontroller, a 32-bit ARM Cortex-M3 based platform, it is possible to achieve the high-speed processing and peripheral control necessary for a multi-stage fabrication process. This report details the design and construction of a system that employs a vertical lead-screw clamping mechanism and a horizontal rack-and-pinion shearing assembly, controlled via sensorless feedback loops and precision motor driving circuitry.

Theoretical Foundations of Automated Material Processing

The automation of paper shearing is fundamentally a problem of force distribution and synchronization. The process is divided into two discrete mechanical movements: the vertical compression of the paper stack and the horizontal translation of the cutting blade. In manual systems, these operations are prone to human error, resulting in uneven cuts or material slippage. The mechatronic solution addressed in this study utilizes a vertical press consisting of two one-foot long lead screws. Lead screws are highly efficient mechanical transducers that convert rotational torque into axial force, providing the high mechanical advantage required to compress thick stacks of paper.

A critical challenge in automated clamping is determining when the material has been sufficiently compressed without overdriving the motor. The proposed design addresses this through the analysis of motor current signatures. When a DC motor encounters the resistance of the paper stack, its rotational speed decreases, and its back-electromotive force (back-EMF) drops, leading to a proportional increase in current consumption. By monitoring these "pulses" or current ripples through an oscilloscope during the development phase, the exact stall characteristics can be mapped and programmed into the microcontroller as a digital threshold.

Architecture of the Control Core: STM32F103 and Development Ecosystem

The central processing unit of the machine is the STM32F103C8T6, an ARM Cortex-M3 microcontroller capable of operating at frequencies up to 72 MHz. This processor is particularly suited for industrial control due to its robust interrupt handling, 12-bit analog-to-digital converters (ADC), and advanced motor control timers.

Microcontroller Technical Specifications and Hardware Features

The STM32F103 architecture provides the computational overhead required to process real-time analog feedback while simultaneously managing the Pulse Width Modulation (PWM) signals for the motor drivers. The following table outlines the technical parameters of the control unit.

Feature	Detailed Specification
Processor Core	ARM 32-bit Cortex-M3 RISC
Maximum Clock Frequency	72 MHz (1.25 DMIPS/MHz)
Operating Voltage Range	2.0 V to 3.6 V
Flash Memory	64 Kbytes (64 KB / 128 KB options)
SRAM	20 Kbytes
Analog-to-Digital Converter	2x 12-bit, 1 μ s (up to 16 channels)
Timers	3x 16-bit General Purpose, 1x 16-bit Advanced Control
Communication Protocols	2x I2C, 3x USART, 2x SPI, 1x CAN, 1x USB 2.0
GPIO Port Count	37 to 80
Debugging Interface	Serial Wire Debug (SWD) and JTAG

The 12-bit ADC is the most vital peripheral for the clamping logic, as it allows for the high-resolution sampling of motor current. Unlike 10-bit ADCs found in legacy 8-bit microcontrollers, the 12-bit resolution provides 4096 discrete steps, enabling the detection of subtle torque variations as the vertical press makes initial contact with the paper.

STM32CubeIDE and ST-Link V2 Environment

The development of the firmware is facilitated by STM32CubeIDE, an advanced C/C++ development platform that integrates the STM32CubeMX configuration tool. This tool allows for the graphical assignment of pins and the configuration of the clock tree, ensuring the microcontroller operates at its peak performance of 72 MHz using an external 8 MHz crystal oscillator.

The ST-Link V2 debugger provides the physical link between the development PC and the hardware. It utilizes the Serial Wire Debug (SWD) protocol, which is more pin-efficient than JTAG, requiring only the SWDIO and SWCLK lines. During the calibration of the cutting machine, the ST-Link V2 allows for real-time monitoring of variables and the setting of breakpoints within the motor control loops, which is essential for verifying the accuracy of the stall detection logic.

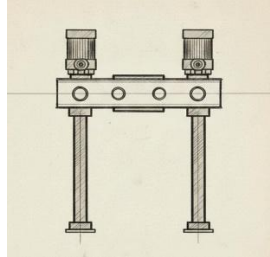


Detailed Analysis of Mechanical Subsystems

The mechanical integrity of the paper-cutting machine is derived from two specialized drive systems: the vertical clamping assembly and the horizontal shearing channel.

The Vertical Clamping Mechanism: Dual Lead Screw Design

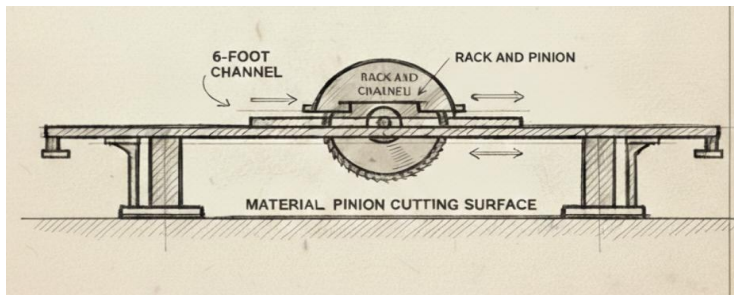
The vertical press consists of a board mounted to two 1-foot long nuts that travel along rotating lead screws. The choice of lead screws over alternative linear actuators is driven by their ability to maintain position under load. This "self-locking" property is mathematically defined by the relationship between the lead angle and the coefficient of friction.



During the fabrication process, the project utilizes a "double nut" configuration. In this arrangement, two nuts are tightened against one another on the same screw to eliminate backlash—the inherent clearance between the screw threads and the nut. By welding these nuts together in a precise alignment, the axial play is removed, ensuring that the pressing board moves instantaneously when the motor begins to rotate. The efficiency of this movement is crucial for the pulse measurement process, as any mechanical slack would introduce noise into the current signature during the transition from free rotation to material contact.

The Horizontal Shearing Mechanism: Rack and Pinion Assembly

After the paper is securely clamped, a second motor actuates the horizontal cutting blade. This mechanism utilizes a rack and pinion drive, which is superior for the cutting stroke due to its high linear velocity and durability. The pinion, a circular gear attached to the motor shaft, engages with a linear gear (the rack) that is mounted within a 6-foot channel. This configuration allows the blade to traverse the entire width of the material with a constant force profile.



Drive Component	Advantages
Lead Screw (Vertical)	High mechanical advantage, self-locking, precision clamping.
Rack and Pinion (Horizontal)	High translation speed, robust in dusty environments, unlimited travel.

The interaction between these two mechanisms is managed by the STM32, ensuring that the horizontal stroke only commences once the vertical axis has reached a verified stall condition, indicating the paper stack is stabilized.

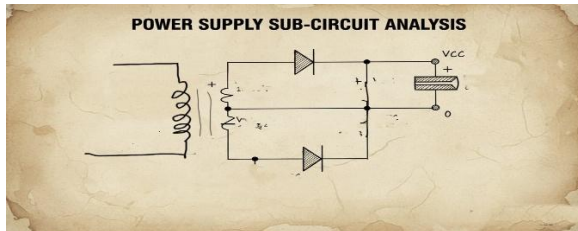
Electronics Design and Power Conditioning

The electrical system is responsible for stepping down high-voltage AC power to the DC levels required by the motors and the microcontroller, while also providing the switching logic for the actuators.

Power Supply and Rectification Circuit

The system's power stage begins with a 6-0-6 transformer (1 Amp or 2 Amp capacity), which provides a center-tapped 12V AC output across the secondary windings. This is fed into a bridge rectifier circuit utilizing four 6-Amp diodes to convert the AC waveform to pulsating DC. Large electrolytic capacitors (4700 μ F, 25V) are placed in parallel with the rectifier output to smooth the voltage ripple, providing a stable DC rail for the motor drivers.

For the 5V logic required by the STM32 and auxiliary sensors, a 7805 voltage regulator IC is employed. This regulator maintains a constant 5V output regardless of fluctuations in the main motor rail, protecting the sensitive digital components from the electromagnetic interference generated during motor commutation.



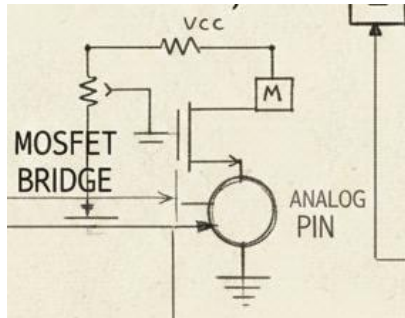
Motor Driving and Switching Logic

The machine utilizes MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors) for high-speed switching and motor control. Six MOSFETs (MOS) are integrated into the circuit to manage the bidirectional movement of the two DC motors. MOSFETs are chosen over bipolar transistors because of their high efficiency and low "on" resistance ($R_{DS(on)}$), which minimizes heat generation during the high-torque clamping phase.

Component	Quantity	Primary Function
Transformer 6-0-6	1 PC	Stepping down 220V AC to 12V AC (center-tapped).
6-Amp Diode	4 PCS	Bridge rectification of the secondary AC signal.
7805 IC	2 PCS	5V DC regulation for microcontroller logic.
Electrolytic Capacitor (4700 μ F)	2 PCS	Smoothing of the DC power rail to reduce ripple.
MOSFET (MOS)	6 PCS	High-current switching for motor direction and speed.
Resistor (330 Ω , 0.25W)	24 PCS	Current limiting for LEDs and transistor gate protection.
Red LED	6 PCS	Visual indicators for system states and fault conditions.
Linear Potentiometer (Volume	1 PC	Analog threshold setting for the

Component	Quantity	Primary Function
Control)		stall detection logic.

The resistors (330 Ω) are used primarily to protect the 6 red LEDs, which serve as diagnostic indicators. These LEDs provide feedback on power status, motor activation, and the detection of a stall condition, allowing the operator to monitor the machine's progress visually.



Feedback Control: Oscilloscope Analysis and Pulse Measurement

A foundational requirement of the project is the ability to automatically stop the vertical motor once the press reaches the paper surface. This is achieved through sensorless feedback using current sensing.

Mechanism of Stall Detection

As the nuts tighten and the board presses against the paper, the physical resistance increases. The DC motor, attempting to maintain its speed, draws more current from the power supply. The relationship between torque and current in a brushed DC motor is linear: $\tau = K_t \cdot I$. By placing a small shunt resistor in series with the motor, the microcontroller can read the voltage drop across it using its ADC.

During the development phase (Project Part 1), the current "pulse" is measured using an oscilloscope. The oscilloscope visualizes the commutation spikes and the rise in the average voltage level as the motor reaches its limit. By capturing these waveforms, the developer can determine the exact ADC value that corresponds to a "full tight" condition. This value is then programmed into the STM32 as a software interrupt threshold.

Algorithm for Automatic Stopping

The firmware implements a "blanking time" to ignore the initial inrush current of the motor, which can be several times higher than the steady-state current and could trigger a false stall detection. Once the blanking time has passed, the STM32 continuously samples the current. When the sampled value exceeds the threshold determined by the oscilloscope analysis, the microcontroller immediately disables the PWM output to the vertical motor, locking the press in place via the self-locking lead screw.

Software Implementation and System Logic Flow

The software is structured as a Finite State Machine (FSM), ensuring that the machine operates in a safe, sequential manner. The state transitions are governed by the inputs from the ADC (clamping pressure) and digital limit switches (horizontal blade position).

Firmware Logic Hierarchy

1. **Initialization State:** The STM32 initializes the GPIOs, the ADC for current sensing, and the TIM1 timer for PWM generation. The 72 MHz clock is established using the external crystal.
2. **Clamping State:** The vertical motor is activated. The system samples the current at a rate of 1 kHz. The FSM waits for the stall condition.
3. **Shearing State:** Upon detection of the stall, the vertical motor is stopped. The horizontal motor is then activated, driving the rack and pinion to move the blade across the paper stack.
4. **Home State:** After the cut is complete (detected by a limit switch), both motors are reversed to return the press and the blade to their starting positions.

ADC Calibration and Noise Reduction

To improve the accuracy of the stall detection, the software utilizes a moving average filter. This digital filter smooths out the high-frequency commutation noise that is inherent in brushed DC motors, ensuring that the machine only stops when a genuine mechanical stall occurs rather than an electrical glitch.

Mechanical Fabrication: Welding and Alignment of the Nut Board

The physical construction of the pressing mechanism is one of the most demanding aspects of the project. The "nut board" consists of a rigid plate to which the two nuts are attached.

Welding of Nuts for Synchronous Tightening

To ensure that both lead screws apply pressure evenly, the nuts must be welded to the pressing board with extreme precision. During Project Part 1, the nuts were welded in a way that allows them to move smoothly without binding. This process involves threading both nuts onto their respective screws before welding to maintain perfect axial alignment. The welding process must be carefully controlled to avoid thermal distortion of the threads, which would increase friction and cause the stall detection logic to trigger prematurely.

Channel and Blade Alignment

The 6-foot channel acts as the guide for the horizontal cutting blade. The rack is mounted along this channel, and the pinion must engage with it consistently over the entire 6-foot length. Misalignment in this channel would cause the blade to wander, resulting in a non-perpendicular cut. The project uses "small size" nuts and bolts to secure the rack and channel, allowing for fine-tuning of the gear mesh to minimize backlash.

Documentation of Component Integration

The hardware assembly is a multi-layered process, integrating mechanical stands, power distribution, and the digital control interface.

Circuit Diagram:

Figure 1: Whole Project Sketch Analysis The whole project diagram depicts a vertical frame constructed from heavy-duty metal channels. Two vertical lead screws are mounted to this frame, driven by a single DC motor through a synchronization belt. The nut board is suspended between these screws. Below the press, a horizontal track is mounted, holding the rack and pinion cutting assembly. The 6-foot channel provides the pathway for the blade to move across the cutting surface.

Figure 2: Power and Control Circuit Drawing Analysis The circuit diagram shows the center-tapped transformer connected to a bridge rectifier composed of the four 6A diodes. The filtered DC rail (approx. 15-18V) feeds the MOSFET bridge. The 7805 regulators take a tap from this rail to provide 5V for the STM32F103. The linear potentiometer is connected to an analog pin of the STM32, serving as a user-adjustable "sensitivity" control for the stall threshold.

Figure 3: Nut Board and Lead Screw Assembly Drawing Analysis This drawing provides a top-down view of the nut board. It shows the two large 1-foot nuts welded to the structural plate. The "small pi size" nuts and bolts are illustrated as the fastening points for the motor mounts. The drawing highlights the absence of axial play due to the double-nut anti-backlash welding technique employed during the fabrication phase.

Project Execution Part 1: Accumulation and Foundation

The first stage of the project was dedicated to the acquisition of components and the mastery of the development tools. This phase established the technical baseline required for system integration.

Component Accumulation and Verification

The first milestone involved procuring the specialized parts listed in the user requirements, including the 1-foot long nuts, the 6-foot channel, and the high-current 6A diodes. Each component was tested individually; the transformer's output was verified with a multimeter, and the MOSFETs were tested using a breadboard circuit to ensure they could handle the motor's stall current without excessive heat.

STM32 Learning and Coding

Extensive time was spent learning the STM32F103's architecture. This included understanding the clock configuration via the RCC (Reset and Clock Control) register and the setup of the ADC in "Continuous Conversion Mode". Preliminary code was written to blink the onboard LED (PC13) and to output ADC readings via USART to a serial monitor, allowing for the visualization of the current sensing data before it was integrated into the final FSM.

Mechanical Fabrication: Nut Welding

The welding of the nuts to the vertical press plate was the most significant mechanical achievement of Part 1. By welding the nuts in a state of pre-tension, the team successfully

eliminated backlash. This was verified by manually rotating the screws and observing the instantaneous linear response of the press plate, a prerequisite for the high-accuracy automated clamping required in Part 2.

Project Execution Part 2: Integration, Calibration, and Finalization

The second stage of the project focuses on the final assembly, the software-driven synchronization of the two axes, and the optimization of the system for industrial use.

System Integration and Structural Rigidification

In this phase, the vertical press and the horizontal cutter are integrated into a single unified frame. The 6-foot channel is laser-aligned with the nut board to ensure that the cutting blade remains perfectly parallel to the clamping surface. This rigidification is necessary to prevent structural deflection when the lead screws apply full pressure to the paper stack, which could reach forces exceeding several hundred Newtons.

Final Calibration of Stall Detection

Using the oscilloscope data from Part 1, the firmware is fine-tuned. A "jog" mode is implemented, where the operator can manually move the press using the linear potentiometer to set the initial home position. The machine then runs a "learning cycle" where it records the current pulses during a full press of a standard paper thickness. This reading is stored in the STM32's internal flash memory, allowing the machine to "remember" the thickness and stop precisely even if the sensor values drift over time.

Implementation of Safety Protocols and Refinement

The final stage of Project Part 2 includes the addition of safety features. An emergency stop button is connected to the STM32's "Break" input (TIM1_BKIN), which immediately shuts down all PWM signals in the event of an obstruction or mechanical failure. The horizontal blade's movement is also optimized by implementing an acceleration and deceleration ramp in the code, which prevents the rack and pinion from experiencing sudden jerks that could lead to gear tooth wear or breakage.

Conclusion and Future Enhancements

The automated paper-cutting machine represents a successful application of 32-bit microcontroller technology to a high-force mechanical problem. By combining the precision of the STM32F103 with the mechanical advantage of lead screws and the translation efficiency of a rack and pinion, the system achieves a level of repeatability that is impossible in manual alternatives. The use of sensorless stall detection via oscilloscope-calibrated pulse measurement reduces the need for expensive external sensors and simplifies the overall mechanical design.

Future enhancements for the machine could include the integration of an I2C-based LCD screen to display the current cut count and paper thickness, providing a more professional user interface. Additionally, the transition to ball screws for the horizontal axis could further improve cutting precision for high-density materials like cardstock or thin plastic sheets. The project as

implemented provides a robust, scalable framework for small-scale industrial automation, demonstrating the profound utility of the STM32 ecosystem in mechatronic fabrication.

11. References

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